P15

The Fourier reconstruction of a head section

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Abstract

The Fourier reconstruction may be viewed simply in the spatial domain as the sum of each line integral times a weighting function of the distance from the line to the point of reconstruction. A modified weighting function simultaneously achieves accuracy, simplicity, low computation time, as well as low sensitivity to noise. Using a simulated phantom, the authors compare the Fourier algorithm and a search algorithm. The search algorithm required 12 iterations to obtain a reconstruction of accuracy and resolution comparable to that of the Fourier reconstruction, and was more sensitive to noise. To speed the search algorithm by using fewer interactions leaves decreased resolution in the region just inside the skull which could mask a subdural hematoma.

Real-World Impact & Application

The Shepp-Logan Phantom & Modern Computed Tomography: Shepp & Logan developed the mathematical Fourier reconstruction algorithms for 2D and 3D cross-sectional head scans. In doing so, they created the iconic 'Shepp-Logan Phantom'—a synthetic digital model of human cranial tissue that remains the universal gold standard used to test, calibrate, and benchmark virtually every CT and PET scanner in hospitals worldwide.